Cut-Generated Information Transport and Complete Observer Verification

A finite mathematical framework for quantum-gravity research

16 October 2026 | Version 1.0

Abstract

We give a complete representation and verification procedure for information transport in a declared finite cut/operator framework. Successive binary projective cuts generate an isometric coherent history, retaining the full input operator algebra on a known image subspace. A positive quadratic form of rank r has a minimum r-channel weighted amplitude record, with exact descent of form-preserving linear operations. We construct n^2 calibrated quadratic readings at rational circular observer positions that reconstruct every Hermitian observable on n complex channels. This count is minimal for fixed scalar readings of an unrestricted Hermitian observable. The same family gives a necessary-and-sufficient finite certificate for all-input linear invariant transport, an explicit decoder for positive metrics, and a deterministic reconstruction-error bound. An inherited two-channel thermodynamic form shows exactly how a circular probe detects information missed by axis readings after classical branch extraction. The manuscript completes this finite mathematical verification problem and integrates standard dilation, square completion and polarization in one native exact implementation. The resulting interface is available for quantum-gravity models; it does not by itself select gravitational dynamics, quantize local constraints, or establish physical faster-than-light communication.

1 The problem and the completed contribution

An invariant value, a recoverable information record and a complete verification procedure are different objects. Two sources can have the same quadratic amount while differing in phase or in their responses to a later cut. A successful transport claim must specify which information is retained, which operations are allowed, and which observations can distinguish a correct transport from an incorrect one.

The native research programme formulates these questions through cuts, retained records and comparison-dependent observables. Its quantum-gravity motivation is to relate operational information to geometry without treating an uncalibrated coordinate or operation count as an absolute physical clock. Here we settle a finite mathematical component of that programme: complete quadratic-observer verification of coherent information transport.

The completed chain is

$$\text{declared cuts} \longrightarrow \text{coherent record} \longrightarrow \text{complete observer certificate.} \tag{1}$$

Each arrow has an explicit construction and proof. Assumptions fix the meaning of verification; the identity being tested is not among those assumptions. Circular observer geometry provides a separating family of readings, not a circular logical justification of the conclusion.

The ingredients have established mathematical precedents. Coherent measurement dilation and recovery on an isometry's image are standard [1]. Weighted square completion and polarization are standard linear algebra. Informationally complete measurements are a developed subject [2]; restrictions such as known low rank alter the relevant reconstruction problem [3]. Our contribution is a source-bound integration of these ingredients in the native cut calculus, including an explicit rational observer family, minimum-count proof, transport test, decoder and reproducible exact certificates. We do not claim tomography or measurement dilation as new physical discoveries.

2 Observation contract and notation

Let V be an n-dimensional positive complex carrier with specified adjoint and calibrated coordinates. Executable examples use the exact field $\mathbb{Q} + \mathrm{i}\mathbb{Q}$, with $\mathrm{i}^2 = -1$, represented by the unchanged native cut-scalar engine. General statements are finite-dimensional statements over $\mathbb{C}$. The carrier is an admitted representation of the cut algebra; this paper does not derive it, physical spacetime, or all historical UGD arithmetic from an empty set of premises.

A cut is an orthogonal projection P, with complement $Q = \mathbb{I} - P$. The selected projective implementation has branch operators P, Q. A cut predicate alone does not uniquely select a physical instrument. Likewise, the ordered cut schedule is protocol data; no gravitational field equation is inferred merely from its existence.

A quadratic observable is $H \in \mathrm{Herm}(n)$ with calibrated reading

$$q_H(x) = x^\dagger H x. \tag{2}$$

Hermiticity, quadratic dependence, dimension and calibration are explicit hypotheses. Distinct probe preparations or reproducible expectation values are permitted. The observations below are not simultaneous exact measurements of a single unknown quantum specimen. In a statistical implementation each expectation must be estimated, and uncertainty enters the error bound in Section 5.

At a stipulated uncut level the accessible algebra may be only $\mathbb{C}\mathbb{I}$. Every complex-linear unital map acts identically there, since $T(a\mathbb{I}) = a\mathbb{I}$. This is zero observable distinction within that algebra, not zero geometric dimension or a proof that inaccessible change cannot occur. A nontrivial cut instead generates $\{aP + bQ\} \simeq \mathbb{C} \oplus \mathbb{C}$. Repeating the same ideal cut is idempotent and does not by itself create a clock.

3 Coherent records and the minimum amplitude representation

For a finite ordered cut tree of depth k, let $E_{j,h,b}$ denote the projection for outcome $b \in \{0, 1\}$ at step j, after prefix h. Each node obeys $\sum_b E_{j,h,b}^\dagger E_{j,h,b} = \mathbb{I}$. For a history $h = (h_1, \ldots, h_k)$, define the ordered branch product K_h and record

$$W_k x = \sum_{h \in \{0,1\}^k} |h\rangle \otimes K_h x. \tag{3}$$

The record includes relative branch phases and the final carrier, not just the outcome labels.

Proposition 1 (Complete coherent history)**.** *For every finite declared cut tree, $W_k^\dagger W_k = \mathbb{I}$. If $R_k = W_k W_k^\dagger$, the map $J_k(O) = W_k O W_k^\dagger$ is an injective $*$-isomorphism onto the corner $R_k \,\mathrm{End}(\mathrm{record}) R_k$. It has inverse $B \mapsto W_k^\dagger B W_k$ on that corner.*

Proof. The empty history has $K_\emptyset = \mathbb{I}$. Appending a cut to a prefix gives

$$\sum_b K_h^\dagger E_{j,h,b}^\dagger E_{j,h,b} K_h = K_h^\dagger K_h.$$

Induction yields $\sum_h K_h^\dagger K_h = \mathbb{I}$, proving the isometry. Inserting $W_k^\dagger W_k = \mathbb{I}$ gives $J_k(AB) = J_k(A)J_k(B)$ and $W_k^\dagger J_k(O) W_k = O$; adjoints are preserved. Every corner operator $B = R_k B R_k$ equals $J_k(W_k^\dagger B W_k)$. The corner identity is R_k, not the identity of the larger record carrier. □

No pre-cut Hamiltonian, continuous time or exponential is needed for this construction. A later embedding into continuous dynamics is a separate contract. On amplitudes the decoder $W_k^\dagger$ has norm one. On densities, $W_k^\dagger(W_k \rho W_k^\dagger) W_k = \rho$.

Classicalizing the history produces $\bigoplus_h K_h \rho K_h^\dagger$ and removes cross-history blocks. For a single cut the lost part is $P\rho Q + Q\rho P$. If two inputs have equal first-cut diagonal blocks, every later classical history beginning with that cut is identical for them. The coherent records can nevertheless remain distinct.

Proposition 2 (Minimum weighted amplitude record)**.** *Let $H \geq 0$ have rank r and entries in $\mathbb{Q} + \mathrm{i}\mathbb{Q}$. There are a full-row-rank $r \times n$ matrix L and a positive rational diagonal matrix D such that $H = L^\dagger D L$. Exactly r linear scalar amplitude channels are necessary and sufficient for this positive weighted representation. The record Lx is faithful on $V/\ker H$.*

Proof. For a nonzero positive residual, choose a positive diagonal pivot a and divide its row by a to obtain ℓ. Subtraction of $a\ell^\dagger\ell$ leaves the positive Schur complement and reduces rank by one. A positive matrix with every diagonal zero is zero, since each two-coordinate positive restriction has zero off-diagonal entry. Exact iteration gives r independent rows and positive rational pivots. Conversely, a sum of m positive weighted rank-one forms has rank at most m. Finally, positivity gives $\ker L = \ker H$. □

The nonnegative summands $D_{jj}|(Lx)_j|^2$ admit a binary Pythagorean synthesis with $r-1$ merges and minimum possible maximum depth $\lceil \log_2 r \rceil$ for $r \geq 1$. These are synthesis counts, not classical bit lengths. Retaining only the summands or their sum loses phase; retaining the linear amplitudes preserves the quotient record.

If $T^\dagger H T = H$, kernel invariance gives a unique descended operator U with $LT = UL$ and $U^\dagger D U = D$. To construct it choose $S = L^\dagger(LL^\dagger)^{-1}$ and set $U = LTS$. Since $LS = \mathbb{I}$ and $T\ker L \subseteq \ker L$, this intertwining identity follows directly. This formula applies for $r > 0$; the zero form has the empty quotient record. Additional operations that do not preserve H can require a larger record; the previously proved operation-closure criterion computes the span of rows LT_w over allowed finite words [5, 4].

4 Complete circular-observer verification

Choose $a = 3/5$, $b = 4/5$ and standard coordinate vectors e_i. The observer family is

$$\mathcal{P}_n = \{e_i\}_i \ \cup \ \{ae_i + be_j,\ ae_i + \mathrm{i}be_j : i < j\}. \tag{4}$$

It contains n^2 vectors. Every vector obeys $p^\dagger p = 1$, and $pp^\dagger$ is an orthogonal binary cut. The two-coordinate vectors lie on circles in the real planes spanned by (e_i, e_j) or $(e_i, \mathrm{i}e_j)$. The projectors are separately accessible tests and are not asserted to sum to $\mathbb{I}$ or form a symmetric informationally complete POVM.

Theorem 3 (Complete observer and transport certificate)**.** *Under the declared finite quadratic observation contract:*

1. *The readings $\{q_H(p) : p \in \mathcal{P}_n\}$ reconstruct every $H \in \mathrm{Herm}(n)$. The count n^2 is minimum among fixed scalar quadratic readings for an unrestricted Hermitian observable.*
2. *For any specified linear $T : \mathbb{C}^n \to \mathbb{C}^m$ and specified Hermitian $G_{\mathrm{in}}, G_{\mathrm{out}}$, equality of the input and output form readings on $\mathcal{P}_n$ is equivalent to $T^\dagger G_{\mathrm{out}} T = G_{\mathrm{in}}$ and hence to equality on every input. If both metrics are positive definite, an exact left decoder is $G_{\mathrm{in}}^{-1} T^\dagger G_{\mathrm{out}}$.*
3. *For every coherent cut history W, the transported observers Wp reconstruct the entire image-corner observable $WHW^\dagger$. More generally, they certify $W^\dagger B W = H$ for a specified output observable B.*

Proof. Write

$$d_i = q_H(e_i), \quad r_{ij} = q_H(ae_i + be_j), \quad s_{ij} = q_H(ae_i + \mathrm{i}be_j).$$

For $H_{ij} = u_{ij} + \mathrm{i}v_{ij}$, direct expansion gives

$$25r_{ij} = 9d_i + 16d_j + 24u_{ij}, \tag{5}$$
$$25s_{ij} = 9d_i + 16d_j - 24v_{ij}. \tag{6}$$

Therefore the exact inverse is

$$\begin{aligned} H_{ii} &= d_i, \\ \operatorname{Re} H_{ij} &= \frac{25r_{ij} - 9d_i - 16d_j}{24}, \\ \operatorname{Im} H_{ij} &= \frac{9d_i + 16d_j - 25s_{ij}}{24}. \end{aligned} \tag{7}$$

with $H_{ji} = \overline{H_{ij}}$. It reconstructs a Hermitian matrix from every real reading vector and substituting it back recovers that vector.

The real dimension of $\mathrm{Herm}(n)$ is $n + 2\binom{n}{2} = n^2$. Any fixed m scalar quadratic readings define a real-linear map to $\mathbb{R}^m$; when $m < n^2$, its kernel is nonzero. Thus it cannot separate all observables. Even positivity alone does not remove the obstruction: a nonzero invisible Hermitian direction A makes $c\mathbb{I}$ and $c\mathbb{I} + \epsilon A$ positive with identical readings for sufficiently small nonzero ϵ and $c > 0$.

For transport, the probe differences are the readings of the Hermitian defect $D_T = T^\dagger G_{\mathrm{out}} T - G_{\mathrm{in}}$. The inverse above proves

$$\left[q_{D_T}(p) = 0 \quad \forall p \in \mathcal{P}_n\right] \quad \Longleftrightarrow \quad D_T = 0. \tag{8}$$

If the metrics are positive, $Tx = 0$ implies $x^\dagger G_{\mathrm{in}} x = 0$ and therefore $x = 0$. Multiplication verifies the stated left decoder.

Finally, Proposition 1 gives

$$(Wp)^\dagger (WHW^\dagger)(Wp) = p^\dagger H p.$$

Reconstruction first gives H and then $WHW^\dagger$. Applying the same argument to $W^\dagger B W - H$ proves the compression test. □

Theorem 3 is a finite iff test for a general all-input identity, not a fit to the displayed source example. Removing any one probe produces an explicit invisible direction: apply (7) to a

reading vector with one at the removed position and zero elsewhere. The native certificate constructs these witnesses.

If the observable is known real symmetric, the imaginary probes are unnecessary and the minimum is $n(n+1)/2$. If its trace is known, one axis reading follows from the trace and the other diagonal readings, leaving n^2-1 independent scalar data for an arbitrary trace-fixed Hermitian matrix. Neither count is a universal minimum for pure states, adaptive protocols or arbitrary nonlinear encodings.

On a known history image corner, the count depends on the original n, not the enlarged history dimension. It does not identify arbitrary off-image blocks of an output operator. Likewise, reconstructing a Gram record $xx^\dagger$ fixes a nonzero x only up to one common unit phase. It is not recovery of that phase from classical quadratic readings.

5 Frame covariance, error control and composition

For an invertible calibrated frame change S, transform

$$p' = Sp, \qquad H' = S^{-\dagger} H S^{-1}, \qquad G'_{\mathrm{ref}} = S^{-\dagger} S^{-1}. \tag{9}$$

Then $p'^\dagger G'_{\mathrm{ref}} p' = 1$ and $p'^\dagger H' p' = p^\dagger H p$. The probe family remains circular relative to the transported reference metric. A nonunitary coordinate transformation need not preserve its Euclidean coordinate norm. This common-frame covariance does not license independently unknown calibration changes for different readings.

Proposition 4 (Deterministic error bound)**.** *Suppose every scalar reading in* (4) *has absolute error at most* ϵ*. The reconstructed Hermitian observable satisfies*

$$\|\widehat{H} - H\|_{\mathrm{op}} \le \left(1 + \frac{25(n-1)}{6}\right)\epsilon. \tag{10}$$

The same coefficient bounds an all-input transport defect when every exact probe defect is bounded by ϵ*.*

Proof. Equation (7) bounds each diagonal error by ϵ and each real or imaginary off-diagonal error by $(25+9+16)\epsilon/24 = 25\epsilon/12$. Thus each complex off-diagonal magnitude is at most $25\epsilon/6$. The maximum absolute row sum is bounded by the right side of (10). For a Hermitian matrix, apply the eigenvector equation at a component of maximum magnitude to bound every eigenvalue by that row sum. For a transport defect, conclude $|x^\dagger D_T x| \le \|D_T\|_{\mathrm{op}}\|x\|^2$. □

Calibration error and statistical uncertainty must be included in the reading budget. Small estimated defects do not prove an exactly zero identity. The bound is deterministic and conservative; no optimal sampling claim is made.

Certificates also compose. If $T_1^\dagger G_1 T_1 = G_0$ and $T_2^\dagger G_2 T_2 = G_1$, substitution proves $(T_2T_1)^\dagger G_2 (T_2T_1) = G_0$. No return to the initial point or external time parameter is required. For semidefinite forms, recovery remains a quotient statement as in Proposition 2.

6 Exact source demonstration and information loss

Use the inherited two-channel thermo form and its exact square completion [4]:

$$H = \begin{pmatrix} 2 & 1 \\ 1 & 5 \end{pmatrix}, \qquad L = \begin{pmatrix} 1 & 1/2 \\ 0 & 1 \end{pmatrix}, \qquad D = \mathrm{diag}(2, 9/2), \qquad H = L^\dagger D L. \tag{11}$$

Two independent amplitude channels are necessary and sufficient. One binary Pythagorean merge evaluates the quadratic total.

For $P = \mathrm{diag}(1, 0)$, $Q = \mathbb{I} - P$, use $Wx = |0\rangle Qx + |1\rangle Px$. The full observable $J(H) = WHW^\dagger$ has exact pullback H. In contrast, copying the same H onto both classical branches gives

$$W^\dagger(\mathbb{I}_{\mathrm{label}} \otimes H)W = PHP + QHQ = H - [P, [P, H]]. \tag{12}$$

Its equality to H is equivalent to $[P, H] = 0$, as sandwiching the lost part with P, Q isolates both cross blocks.

Table 1: The four calibrated readings. Coherent transport preserves the source column. Classical branch extraction retains the axes but loses the real cross term.

Observer position	Source H	Fixed branch readout	Difference
$(1, 0)$	2	2	0
$(0, 1)$	5	5	0
$(3/5, 4/5)$	122/25	98/25	$-24/25$
$(3/5, 4\mathrm{i}/5)$	98/25	98/25	0

The inverse reconstructs the lost defect as $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$. The coherent record instead passes all four tests. A complex positive fixture

$$H_c = \begin{pmatrix} 2 & 1 + 2\mathrm{i} \\ 1 - 2\mathrm{i} & 5 \end{pmatrix}$$

has readings $(2, 5, 122/25, 2)$ and reconstructs both cross components exactly. Real-only probes cannot distinguish the Gram records of $(3/5, 4\mathrm{i}/5)$ and $(3/5, -4\mathrm{i}/5)$, while the imaginary circular probe does. Multiplication by a common unit phase changes none of these quadratic readings.

7 What the certificate implies for geometry

The construction supplies a rigorous interface for a proposed information–geometry model: specify its carrier, accessible observables, cuts, retained record and transport; then apply the complete family to its claimed invariant identities. A nonzero reconstructed defect identifies exactly which quadratic information fails to match. This is stronger than checking one radius or a few axis values.

Norm preservation alone does not select curvature. For example, let

$$U = \begin{pmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{pmatrix}, \qquad V = \mathrm{diag}(1, \mathrm{i}), \qquad K = V^\dagger U^\dagger V U. \tag{13}$$

Both $\mathbb{I}$ and this nontrivial loop preserve every Euclidean norm, whereas $K^\dagger P K - P \neq 0$. The complete observer family detects the changed cut observable. Interpreting an algebraic loop as physical spacetime curvature requires an additional geometric and dynamical identification. Also, the nonlinear map $x \mapsto \|x\| e_1$ preserves norms while erasing direction. The linearity hypothesis in the decoder theorem is substantive.

The existing programme has separate geometric and physical results. Its finite hinge and matter–geometry packets provide selected geometric realizations and joint interactions; the local coframe construction has classical constraint closure in a specified $1 + 1$ model [6, 7]. None of these changes the hypotheses of the present theorem. In particular, classical constraint

closure is not a proof of quantum anomaly cancellation or of a positive physical sector in $3+1$ dimensions.

The signalling audit is similarly separate [8]. Local trace-preserving quantum operations give zero remote contrast. The inherited nonlinear branch-sensitive extension yields a conditional finite contrast $1/10$, but requires an additional physical state-law assumption and has no calibrated faster-than-light arrival comparison. Circular verification does not discharge those obligations. No gravitational experiment or physical transmission-speed measurement is reported here.

The completed mathematical result therefore closes this manuscript's representation and verification problem. To close a physical quantum-gravity construction would additionally require a justified matter/geometry law, its physical quantum constraints and domains, appropriate limiting behaviour, and calibrated distinguishing observations. These are separate research claims rather than omitted steps in the finite reconstruction proof.

8 Reproducibility and certification scope

The executable packet is `papers/native-cut-generated-history/` in the Publications repository [4]. Python 3.12 performs orchestration, source hashing and I/O; the unchanged native JavaScript scalar/matrix and Paninian rewrite engines perform scientific arithmetic and symbolic replay. There is no numerical fitting or floating-point scientific calculation in the new observer certificate.

Table 2: Executed certificate counts. Counts describe verification work, not the number of independently novel theorems.

Packet	Exact checks	Symbolic replays	Negative controls
CHD-1/2 retained	348	8	36
CHD-3 addition	246	4	53
Combined CHD	594	12	89

The new execution checks complete Hermitian bases in dimensions $1, 2, 3, 5$, all omitted-probe dual witnesses for those families, source reconstruction, coherent-history compression, valid and invalid transports, common-frame covariance, phase blindness, a nontrivial isometric loop and every vertex of the two-channel reading-error box. Four symbolic certificates cover the real and imaginary polarization expansions and their rational inverses. A corrupted witness is rejected. Every earlier CHD-1/2 certificate entry and result remains a prefix or unchanged subrecord.

General dimension, arbitrary-depth and error-bound claims have the written proofs above. Finite fixtures do not certify their universal quantifiers, and native rewrite replay is not a general proof-assistant formalization of the manuscript. The full research runner has 23 stages, all passing at the recorded source revision; GitHub CI independently passed the same research replay and its inherited action gate.

```
python3 papers/native-cut-generated-history/run.py -check
python3 scripts/replay_quantum_geometry.py
```

Evidence revision: `ac291f7aefea5b6f3d0eaf7f7679e83625341285`. The combined certificate SHA-256 is

```
91cdbd7de3c0b640f7dcd9b3f6022ef509e58cc72675c8d05298ab532c069c4b
```

The accompanying manuscript manifest binds this paper to its local proof sources and certificate bytes. It records provenance, not an additional physical or theorem certificate.

9 Conclusion

We have completed a finite operational verification problem: declared binary cuts construct a recoverable coherent history; a positive quadratic source has a minimum weighted amplitude record; and a minimum calibrated family of circular observer positions reconstructs every quadratic observable and certifies its linear transport on all inputs. The construction includes an explicit decoder, common-frame covariance, a deterministic error bound and an exact source example that exposes information missed by axis-only checks.

Within this scope, the observation contract determines what can be verified, and the separating observer family proves the claimed invariant identity without assuming it. The completed interface can now be used to test a specified quantum-gravity model's information-transport claims. Selecting and validating that physical model remains distinct from the mathematical completion established here.

References

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